

PHYSICAL MODEL OF GAMMA-RAY BURST SPECTRAL EVOLUTION

EDISON LIANG, MASAOKI KUSUNOSE,¹ I. A. SMITH, AND A. CRIDER

Department of Space Physics and Astronomy, MS108, Rice University, Houston, TX 77005-1892

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ABSTRACT

We propose a physical model of gamma-ray burst spectral evolution in which impulsively accelerated nonthermal leptons cool by saturated Compton upscattering of soft photons. This model naturally explains the recently discovered exponential decay of the spectral break energy with photon fluence, the hard-to-soft spectral evolution patterns, and other spectral and temporal properties of gamma-ray bursts.

Subject headings: gamma rays: bursts

1. INTRODUCTION

Since the launch of BATSE, the debate on the origin of gamma-ray bursts (GRBs) has focused on the global astrophysical issues (Fishman & Meegan 1995). However, information contained in GRB temporal profiles and spectral evolutions, although less tractable than information on spatial distributions, still holds the key to the underlying physics of these enigmatic events. Recently Liang & Kargartis (1996, hereafter LK96) discovered a remarkable property of GRB spectral evolution: for bright, long, smooth pulse decays they showed that the spectral break energy E_{pk} , defined as the peak of the νF_ν distribution (where ν is the photon energy in keV and F_ν is the differential energy flux in $\text{ergs keV}^{-1} \text{s}^{-1} \text{cm}^{-2}$; see Schaefer et al. 1992), often decreases exponentially as a function of photon fluence Φ ($= \int dt F_N$ [photon number flux in the BATSE range]).

We propose that this result, together with other properties of GRB spectral evolution, can be explained by a *saturated synchrotron self-Compton* model, in which impulsively accelerated, nonthermal, relativistic leptons (electrons and pairs) repeatedly Compton upscatter self-emitted (radio-IR) synchrotron photons into gamma-ray energies. The Thomson depths τ_T are initially large ($\gg 1$; Zdziarski, Coppi, & Lamb 1990; see also Fenimore et al. 1982 for the thermal case) so that the emerging gamma rays reach kinetic equilibrium with the lepton distribution and the synchrotron soft-photon source is strongly self-absorbed by internal free-free and synchrotron opacities (Melrose 1980). The observed spectral break E_{pk} is related to the low-energy cutoff of the nonthermal lepton distribution, and the anticorrelated evolution of flux and E_{pk} at the beginning of the hard-to-soft evolution (Norris et al. 1986; Ford et al. 1995) can be explained by the combination of Compton cooling and optical thinning of the plasma, possibly due to expansion.

2. EVIDENCE FOR SATURATED COMPTONIZATION

As LK96 proposed, one simple interpretation of the exponential decay of E_{pk} with photon fluence is that a plasma of a fixed number N of leptons with average energy $\langle E \rangle$ cools by emitting radiation with characteristic energy $\langle \nu \rangle$ ($\sim E_{\text{pk}}$) $\propto \langle E \rangle$. Saturated Compton upscattering of soft photons (Rybicki & Lightman 1979) produces such a result, since the emerging photons reach Wien equilibrium with the lepton distribution.

The other evidence for saturated Comptonization comes from the evolution of the GRB spectrum below E_{pk} . Crider et al. (1997) showed that at early times many GRB photon number spectra below E_{pk} exhibit a positive slope that often decreases with time in hard-to-soft pulses. This effect is difficult to explain with spectral models that require a constant or negative asymptotic low-energy slope, such as synchrotron emission (Katz 1994; Tavani 1996) or self-absorption. Yet this behavior is naturally explained by saturated Comptonization with Thomson thinning. Figure 1 shows sample Monte Carlo spectra from the multiple Compton upscattering of a soft photon source by a sphere of relativistic power-law leptons (Smith et al. 1997). Fitting the simulation spectra with the Band GRB function (Band et al. 1993) allows us to determine the asymptotic slopes α and β for the power laws below and above E_{pk} . Above E_{pk} the slope β ($\sim -p$, the particle index, see eq. [1] below) is independent of τ_T , whereas below E_{pk} the slope α decreases with decreasing τ_T . For $\tau_T \gg 1$, α approaches the Wien limit of $+2$, whereas for $\tau_T \ll 1$ the spectrum reduces to a single-scattering Compton spectrum with slope $= -(p+1)/2$ ($= -1.75$ for $p = 2.5$ in Fig. 1a; Rybicki & Lightman 1979; Pozdnyakov, Sobel, & Sunyaev 1983). Comparing Figure 1a with Figures 1 & 2 of Crider et al. (1997) shows the strong resemblance of these GRB continuum evolutions to the evolution of saturated Comptonization with Thomson thinning.

3. NATURE OF THE SOFT PHOTON SOURCE

What is the origin of the soft photons and what causes the rise and decay of the photon flux? We have investigated different soft photon sources (blackbody, nonthermal, internal, external, central, and uniform) and geometries (sphere, slab) to answer these questions. In all cases our simulations give similar Compton upscattered spectra, all of which resemble observed GRB spectra (Smith et al. 1997). Here we show results from our favored model, which generates the soft photons via self-absorbed internal synchrotron radiation.

Consider a population of leptons initially accelerated (in contrast to the continuous injection case of Zdziarski et al. 1990) to form a power law of index p with an initial low-energy Lorentz factor cutoff γ_{min} :

$$N_\gamma \sim \gamma^{-p} \text{ for } \gamma \geq \gamma_{\text{min}}, \quad N_\gamma = 0 \text{ for } \gamma < \gamma_{\text{min}};$$

$$\int N_\gamma d\gamma = N. \quad (1)$$

¹ Permanent address: Kwansei Gakuin University, Vegahara Ichiban-cho, Nishinomiya 662, Japan.

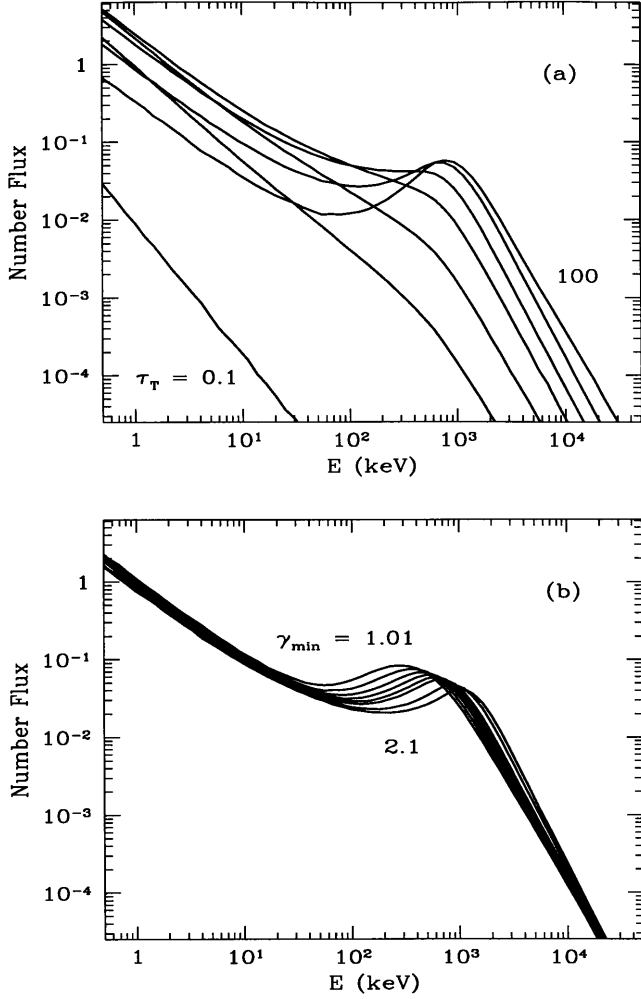


FIG. 1.—Examples of Monte Carlo simulation spectra. A uniform sphere of radius 10^8 cm contains relativistic electrons with Lorentz factors $\gamma_{\min} \leq \gamma \leq 1000$ and power-law index $p = 2.5$. In a uniform magnetic field of 1 G, these electrons emit synchrotron radiation that is multiply scattered up to the gamma-ray region. (a) $\gamma_{\min} = 1.5$ is fixed for all the curves. Left to right: $\tau_T = 0.1, 1, 2, 4, 8, 25$, and 100 . (b) $\tau_T = 25$ is fixed for all the curves. From softest to hardest, $\gamma_{\min} = 1.01, 1.1, 1.2, 1.3, 1.4, 1.5, 1.8$, and 2.1 .

As the leptons cool by Comptonization, $\gamma_{\min}$ decreases with time. Because of the faster cooling of high- γ particles, a high-energy cutoff $\gamma_{\max}$ will develop, which also decreases with time. Our simulations show that this cutoff does not affect the observable spectrum as long as $\gamma_{\max} \gg 1$, which we assume here. For $\gamma < \gamma_{\max}$ the distribution given by equation (1) cools

self-similarly with $p \sim \text{constant}$ (Blumenthal & Gould 1970). In future work, the sharp cutoff at $\gamma_{\min}$ will be replaced by a low-energy Maxwellian peaking at $\gamma_{\min}$ to allow for the thermalization of the lowest energy leptons (Smith et al. 1997). We also assume that any ions present stay fully ionized and couple only weakly to the leptons.

To produce a rising photon flux while the leptons are cooling in a hard-to-soft GRB pulse, we need a soft photon source increasing with time. Coupled with the observed decrease of α , this requirement suggests optical thinning as a cause of the flux rise. This idea in turn implies that the soft photons are self-emitted rather than external, and synchrotron radiation is the most efficient photon production mechanism. In a uniform magnetic field B the leptons radiate synchrotron radiation with a power law of slope $-(p+1)/2$ for frequency $\nu \geq \nu_c$ (critical frequency) and slope $-2/3$ for $\nu \ll \nu_c$ (Rybicki & Lightman 1979). However, the synchrotron spectrum will be strongly self-absorbed at low frequencies as a result of the synchrotron opacity κ_{syn} (Melrose 1980; Rybicki & Lightman 1979) and free-free (ee and e^+e^- bremsstrahlung; Dermer 1984) opacity κ_{ff} (for reasonable densities and B , free-free emission will be negligible compared to synchrotron emission, even when $\kappa_{\text{ff}} > \kappa_{\text{syn}}$ at low frequencies; Harding et al. 1984). In the presence of Compton upscattering, a single scattering by a relativistic lepton is enough to boost the photon energy above the absorption frequency ν_{ab} and allow it to avoid reabsorption. Hence real absorption occurs only below the frequency where $\tau_{\text{ab}}(\nu_{\text{ab}}) = \tau_T$ (instead of the usual $\tau_{\text{ab}}(\nu_{\text{ab}}) = 1$), and we use this result to compute ν_{ab} . Above ν_{ab} the net soft photon injection spectrum is given by the optically thin synchrotron formula, while below ν_{ab} it is given by the usual $+1.5$ slope (Rybicki & Lightman 1979). From this we derive the net synchrotron photon injection rate per lepton available for Comptonization:

$$\epsilon_{\text{syn}}^{\text{inj}} = A \gamma_{\min}^n \tau_T^{-q} B^r, \quad (2)$$

where A, n, q , and r are functions of p . Table 1 lists results for sample p -values for $\kappa_{\text{ff}} \gg \kappa_{\text{syn}}$ (high-density, low- B) and $\kappa_{\text{ff}} \ll \kappa_{\text{syn}}$ (high- B , low-density), respectively.

Table 1 shows that when $\kappa_{\text{ff}} \gg \kappa_{\text{syn}}$, $\epsilon_{\text{syn}}^{\text{inj}}$ varies inversely with τ_T , whereas when $\kappa_{\text{ff}} \ll \kappa_{\text{syn}}$, $\epsilon_{\text{syn}}^{\text{inj}}$ is independent of τ_T . Hence we speculate that extreme hard-to-soft evolution (Norris et al. 1986) may be associated with κ_{ff} -domination and τ_T thinning, whereas tracking evolution (Golenetski et al. 1983; Liang 1992) may be associated with κ_{syn} -domination or a constant τ_T . In the former case the flux rises and falls due to the combined effect of falling $\gamma_{\min}$ and τ_T . In the latter case the flux simply varies with $\gamma_{\min}$ (see eq. [2]).

TABLE 1
SELF-ABSORBED SYNCHROTRON EMISSIVITY

Parameter	$\kappa_{\text{ff}} \ll \kappa_{\text{syn}}$		$\kappa_{\text{ff}} \gg \kappa_{\text{syn}}$	
	$p = 2.5$	$p = 3.5$	$p = 2.5$	$p = 3.5$
A	14.7	0.24	$1.1 N_{45}^{3/16}$	$1.5 \times 10^{-3} N_{45}^{5/16}$
n	1.154	1.667	1.875	3.125
q	0	0	0.563	0.938
r	1.231	1.333	1.750	2.25
$\nu_{\text{ab}}(\text{Hz})$	$2.8 \times 10^{11} B^{9/13} \gamma_{\min}^{6/13}$	$8.1 \times 10^{10} B^{11/15} \gamma_{\min}^{2/3}$	$1.3 \times 10^{12} D$	$1.6 \times 10^{12} D$

NOTE.— $D = N_{45}^{-1/4} \gamma_{\min}^{-1/2}$ and $N_{45} = (\text{total lepton number})/10^{45}$. 10^{45} is the typical number of leptons needed in our model for an extended-halo GRB (~ 150 kpc for the faint BATSE bursts).

4. A SIMPLE ANALYTIC MODEL FOR THE PULSE PROFILE

From Monte Carlo simulations we find $E_{\text{pk}} \propto (\gamma_{\text{min}} - 1)$ for γ_{min} not too close to 1. Hence the empirical result of LK96 can be rewritten as

$$\gamma_{\text{min}} - 1 \approx (\gamma_0 - 1) \exp(-\Phi/\Phi_0), \quad (3)$$

where γ_0 is the peak value of γ_{min} in a pulse and Φ_0 is the decay constant in LK96. Since Compton scattering conserves photons, the soft photon injection rate (eq. [2]) can be related to the BATSE flux $F_N = d\Phi/dt$ via

$$\epsilon_{\text{syn}}^{\text{inj}} = (d\Phi/dt)b\Omega d^2/N, \quad (4)$$

where b = (soft photon injection flux/BATSE photon flux), d = GRB distance, and Ω = emission solid angle. Combining equations (2), (3), and (4), we have

$$[1 + (\gamma_0 - 1) \exp(-\Phi/\Phi_0)]^{-n} (d\Phi/dt) = AN\tau_T^{-q} B^r b^{-1} \Omega^{-1} d^{-2}, \quad (5)$$

which can be integrated to obtain $\Phi(t)$, once we specify $\tau_T(t)$, $B(t)$, and $b(t)$. In principle, $b(t)$ can only be derived using Monte Carlo simulations. However, to illustrate the basic qualitative property of the solution to equation (5), we adopt the following analytic model. Let $b = [1 + (\gamma_0 - 1)^{-1} \exp(\Phi/\Phi_0)]^n$ and $B = \text{constant}$, and let τ_T evolve as a power law in time ($\tau_T^q \propto t^s$; $s > 0$ simulates a Thomson thinning hard-to-soft pulse, while $s = 0$ simulates the decay phase of a tracking pulse). We then have the analytic solution

$$\begin{aligned} \Phi(t) &= \Phi_0 n^{-1} \ln(1 + Kt^{1+s}), \\ F_N(t) &= \Phi_0 n^{-1} K(1+s)t^s/(1 + Kt^{1+s}), \\ \gamma_{\text{min}} - 1 &= (\gamma_0 - 1)(1 + Kt^{1+s})^{-1/n}, \\ b(t) &= (1 + (\gamma_0 - 1)^{-1}(1 + Kt^{1+s})^{1/n})^n, \end{aligned} \quad (6)$$

where K is a constant. The above $b(t)$ is qualitatively consistent with Monte Carlo simulation results. Figure 2 shows Monte Carlo simulation results of GRB pulses patterned after equation (6) for both $s = 0$ and $s = 1$. Figure 3 gives the corresponding spectral evolution snapshots, which resemble the BATSE data of Crider et al. (1997).

Equation (6) shows that at late times, $F_N \sim t^{-1}$, independent of n or s , and $E_{\text{pk}} \propto (\gamma_{\text{min}} - 1) \propto t^{-(1+s)/n}$. Hence

$$F_N \sim E_{\text{pk}}^{n/(1+s)} \quad (7)$$

Such a power-law correlation during pulse decays has been reported (Golenetskii et al. 1983; Kargatis et al. 1995). Also, the above $1/t$ decay profile is consistent with the $1/\omega^2$ power density spectra reported by Shaviv (1996).

Real GRB pulses are of course much more complex than the above analytic model, which is introduced for illustrative purposes only. Yet this simple model seems capable of reproducing the basic profile and spectral evolution of generic long, smooth GRB pulses.

5. ASTROPHYSICAL IMPLICATIONS

$F_{N_{\text{max}}}/\Phi_0$ measures the peak gamma-ray output per lepton, independent of the source distance. It is approximately a few photons s^{-1} for typical bright GRB pulses (LK96). Using equation (2), this empirical result uniquely constrains the magnetic field B , since we can estimate p , γ_{min} , b , and τ_T from spectral modeling. For $p = 2.5$ (typical of GRBs), we find

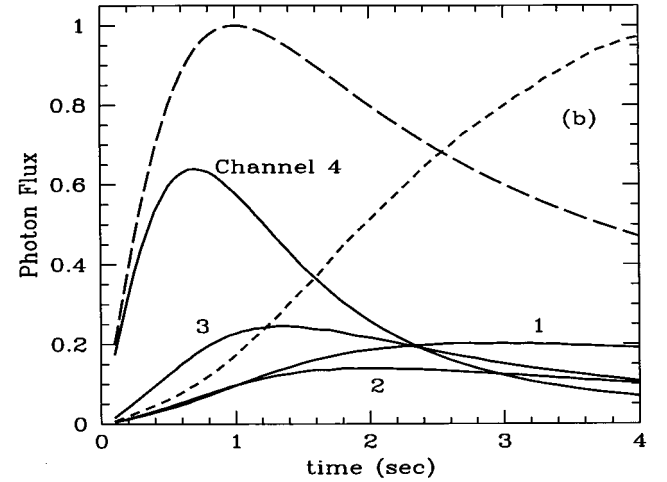
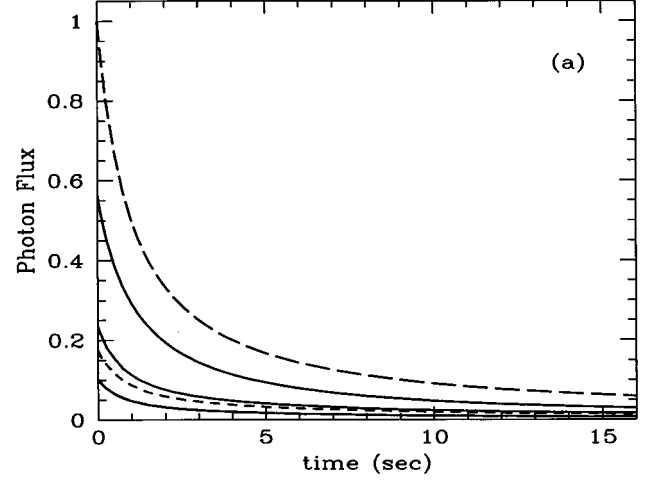


FIG. 2.—Time histories for simulations of (a) a tracking pulse (decay phase only), and (b) a hard-to-soft pulse. For both cases, a uniform sphere of radius 10^{12} cm contains relativistic electrons with Lorentz factors $\gamma_{\text{min}} \leq \gamma \leq 1000$, $\gamma_0 = 4$, and power-law index $p = 2.5$. In a uniform magnetic field of 100 G, these electrons radiate synchrotron radiation that is multiply scattered up to the gamma-ray region. Time histories of γ_{min} , τ_T , and F_N are assumed to follow eq. (6). (a) $s = 0$, $\kappa_{\text{syn}} \gg \kappa_{\text{ff}}$, $K = 1$, $\tau_T = 4$, $n = 1.154$. (b) $s = 1$, $\kappa_{\text{ff}} \gg \kappa_{\text{syn}}$, $K = 1$, $\tau_T(1 \text{ s}) = 4$, $n = 1.875$, $q = 0.56$. Long-dashed curves: 20–1500 keV flux (BATSE channels 1, 2, 3, and 4). Solid curves, top to bottom: 300–1500 keV (BATSE channel 4), 100–300 keV (BATSE channel 3), 50–100 keV (BATSE channel 2), and 20–50 keV (BATSE channel 1). In (a), the channel 1 and 2 time histories are almost indistinguishable. Fluxes have not been convolved with the detector responses. Short-dashed curves: 2–20 keV X-ray fluxes. Note the increase in X-ray excess at late times in case (b). Narrower pulses can be produced if s and K are increased.

$B \sim 0.1\text{--}10$ G for the typical bright GRB pulse. This important result is independent of the source distance.

Another important result of this model is the ratio of the source distance d to the emission region size R . From equation (3) of LK96 with $\langle \nu \rangle \sim \langle E \rangle$, we have

$$d^2 \approx N/(\Omega\Phi_0) = g4\pi R^2\tau_T/(\sigma_T\Omega\Phi_0),$$

where $\sigma_T = 6.65 \times 10^{-25} \text{ cm}^2$ is the Thomson cross section and g is a geometric factor (e.g., $=1$, for a thin spherical shell, $=1/3$ for a solid sphere). Hence

$$d/R = 1.3 \times 10^{12} (g\tau_T/10)^{1/2} [(\Omega/4\pi)(\Phi_0/10)]^{-1/2}, \quad (8)$$

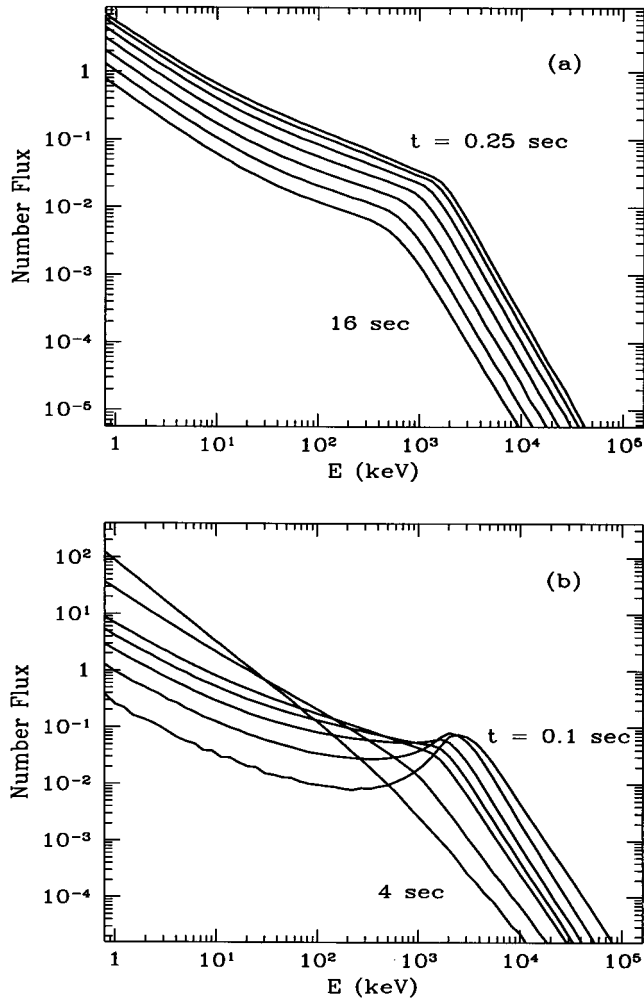


FIG. 3.—Snapshots of the spectra for the simulations shown in Fig. 2. (a) $s = 0$. The spectrum evolves from hardest to softest, with snapshots shown at $t = 0.25, 0.5, 1, 2, 4, 8$, and 16 s. (b) $s = 1$. The spectrum evolves from hardest to softest, with snapshots shown at $t = 0.1, 0.25, 0.5, 0.75, 1, 2$, and 4 s. These spectra are harder than the typical BATSE bursts because of the large γ_0 , chosen here for illustrative purposes only. A smaller γ_0 gives the more typical result that the X-ray upturn at early times lies completely below 20 keV.

independent of the source distance. Equation (8) is compatible with extended Galactic halo distances if $R \sim$ light seconds and $\Phi_0 \sim 0.1$ for the faintest BATSE bursts. Its compatibility with cosmological distances is under investigation.

6. DISCUSSION

We summarize the most important observational implications of this model here.

1. The largest value of α should occur at the beginning of a hard-to-soft pulse (approaching the Wien limit in the hardest cases) and decrease monotonically. Evidence for this behavior has been reported by Crider et al. (1997).

2. There should be an upturn in the soft X-ray intensity (Fig. 1). Some tentative evidence for this effect has been reported by Preece et al. (1996). At late times there should be an increase in soft X-ray excess (Fig. 2). However, depending on the amount of interstellar and circumburst absorption, such an X-ray upturn or excess may not be detectable in every case.

3. For extreme hard-to-soft pulses, if $B \sim$ constant, the flux evolution should be jointly determined by the evolution of α and E_{pk} , whereas for extreme tracking pulses it depends only on E_{pk} (see eq. [2]).

4. At late times the radio–gamma-ray continuum should be produced by unsaturated Comptonization (Figs. 1a, 3b). Initially the radio flux should peak at 10^{10} – 10^{12} Hz (see Table 1). As the leptons cool and optically thin out, the radio peak should move to lower frequencies.

In this paper we have idealized the plasma as quasistatic during Comptonization and cooling only via radiation and ignored pair processes. The decrease of τ_T with time implies that the plasma is probably expanding, in which case adiabatic cooling and time dependence must be included. The inclusion of pair processes (Zdziarski et al. 1990) is not expected to substantially change our results except for some steepening of the photon index β . These issues will be addressed in future papers.

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